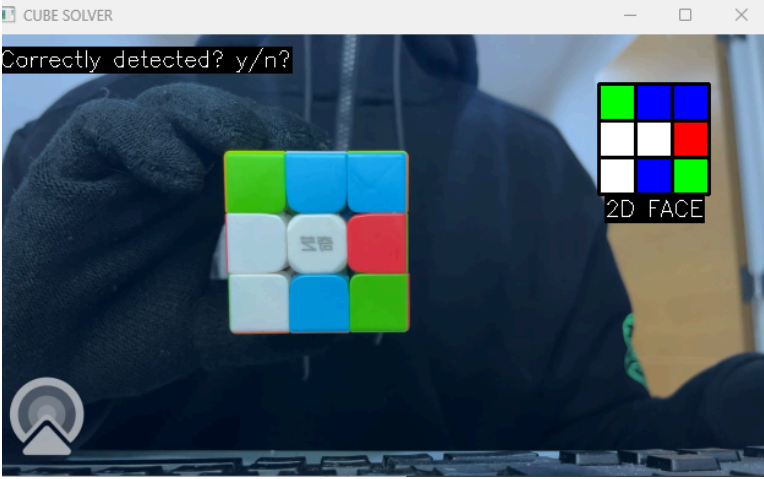
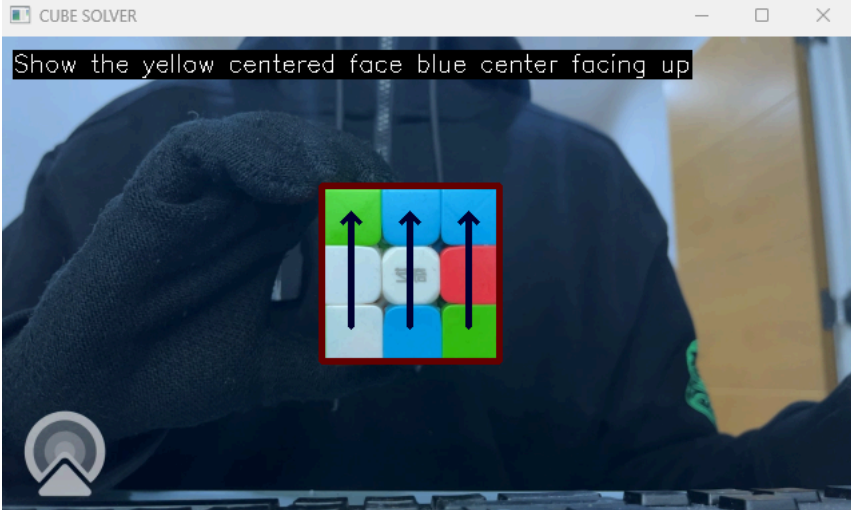
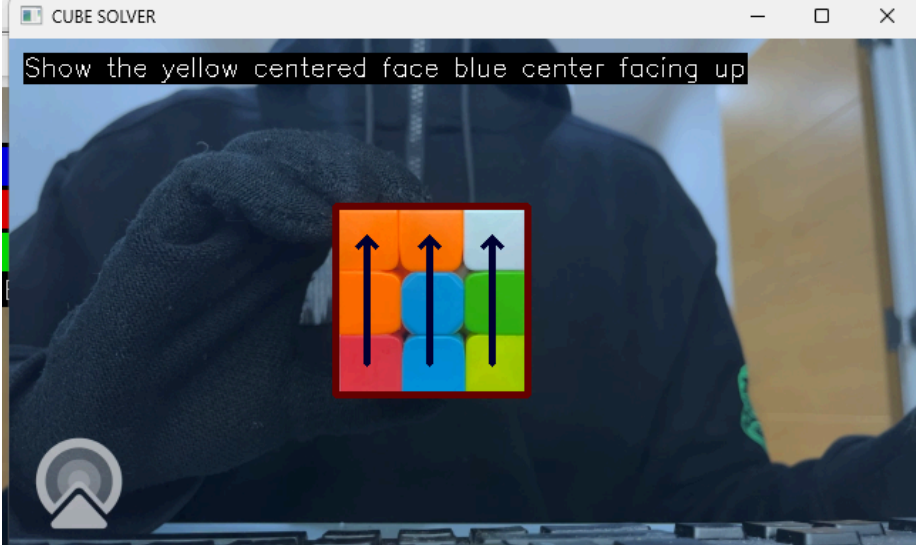
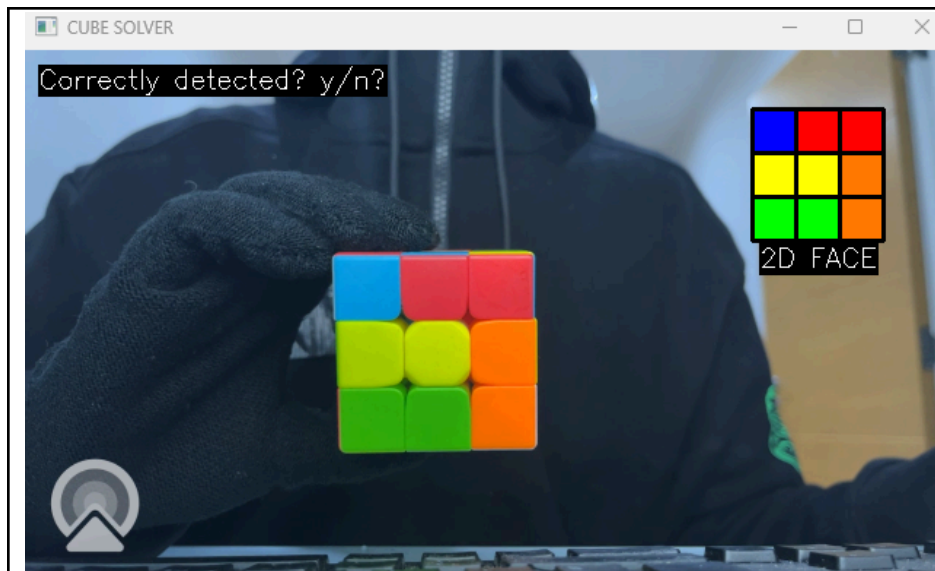
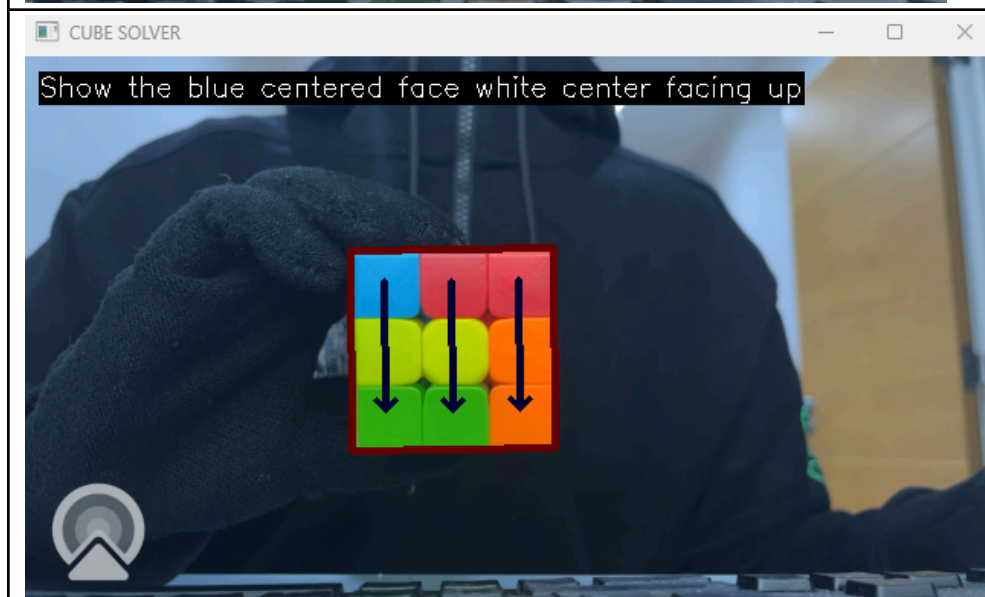


Cube Detection Process

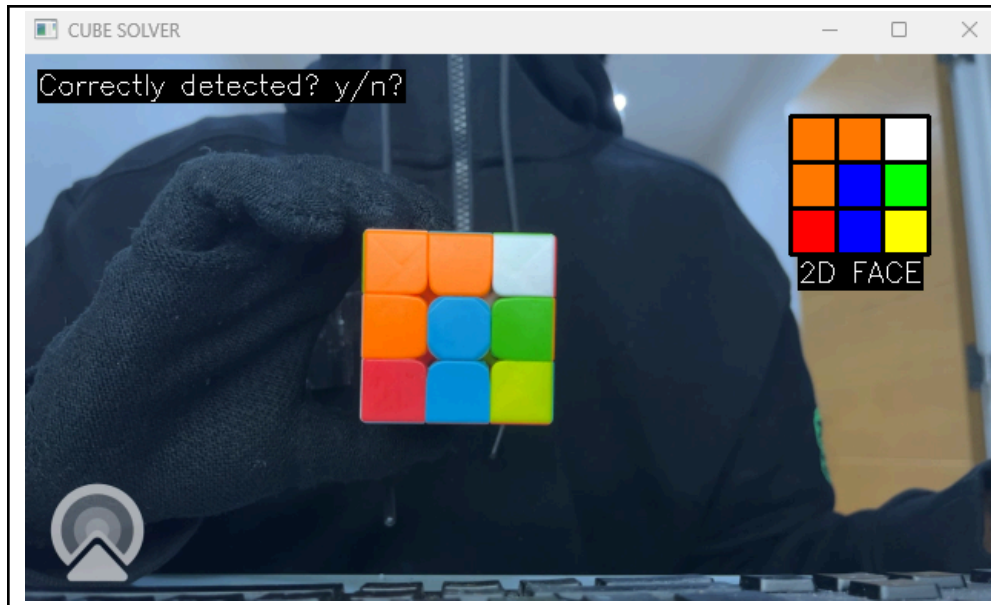
Frame	Description
	1)First I started by presenting the white face to the camera. The application quickly detected the colours of the face and displayed them with minimal delay. This indicates that the colour detection algorithm is efficient and responsive making the overall application more intuitive to use.
	2)Next, I verified that the detected colours matched the colours of the cube itself and were in the same positions. Since the detected colours are right, I pressed the ('Y') key and the steps for displaying the next face are then displayed.
	3) After confirming the white face, I pressed the 'Y' key to proceed. The application displayed instructions for the next face to be detected, which is the yellow face. However, I noticed that detecting the yellow face immediately after the white face is illogical since they are on opposite ends of the cube. This is an area that needs improvement to provide a more intuitive user experience.



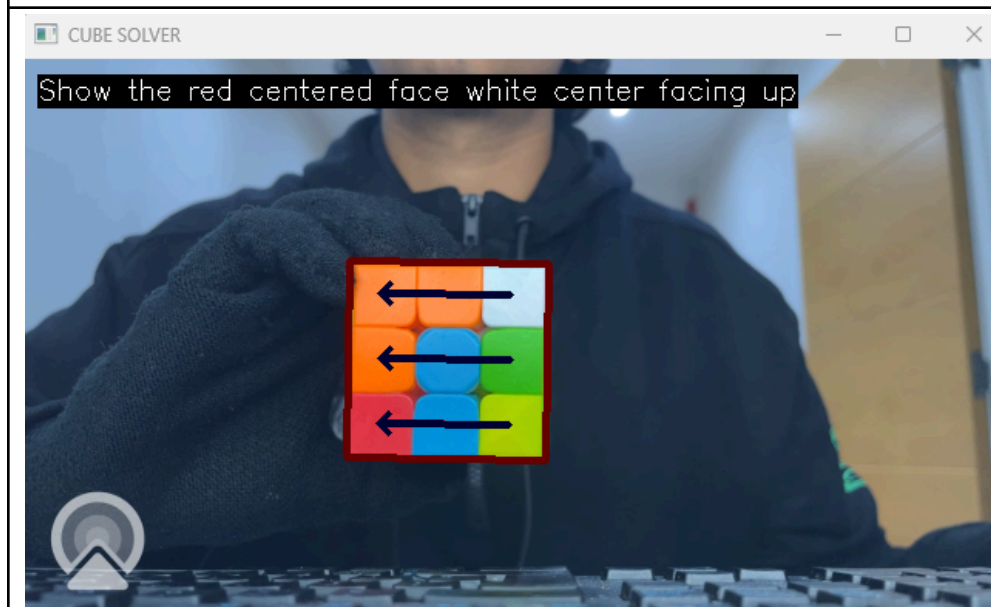
4) Following the augmented arrows, I presented the yellow face to the camera. The application correctly detected and displayed the colours of the yellow face on the 2D representation. This further validates the accuracy and consistency of the colour detection algorithm.



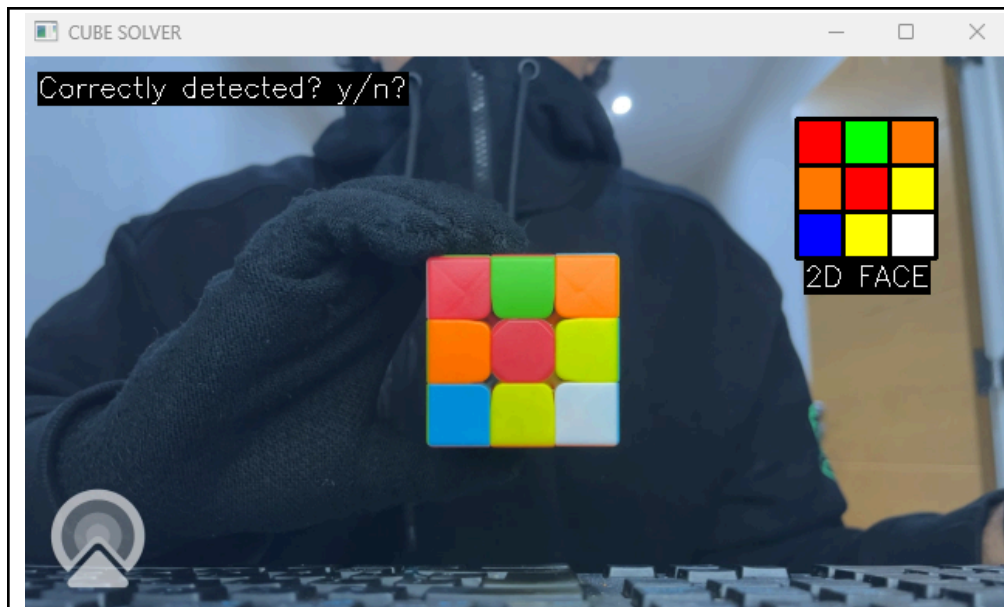
5) The next desired face is the blue face, so the augmented arrows are being drawn in the correct direction to guide the user to displaying the blue face.



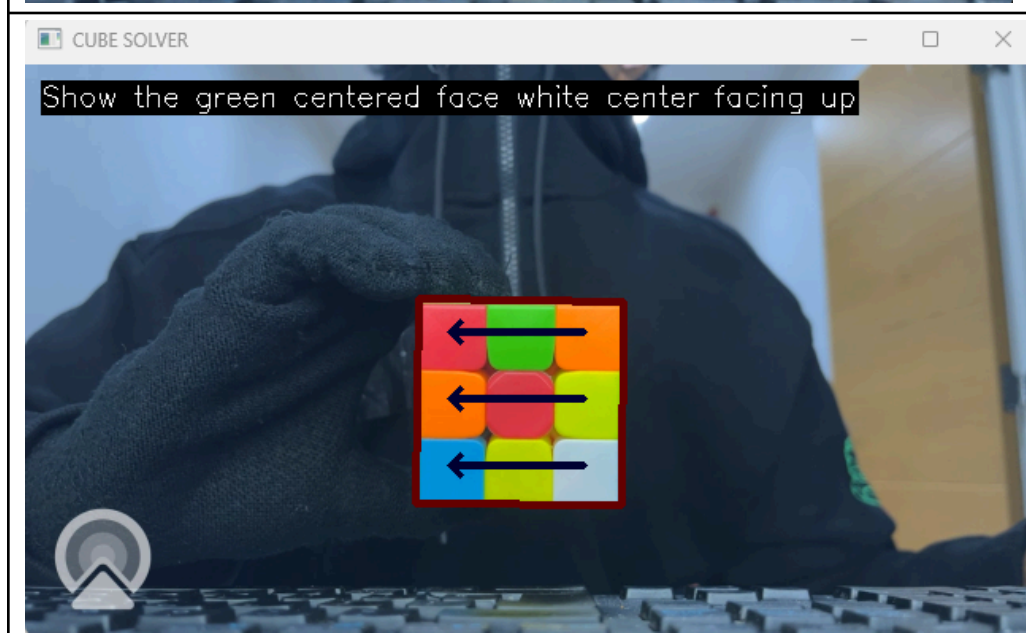
6)The blue face is also being correctly displayed and correctly detected on the 2D face.



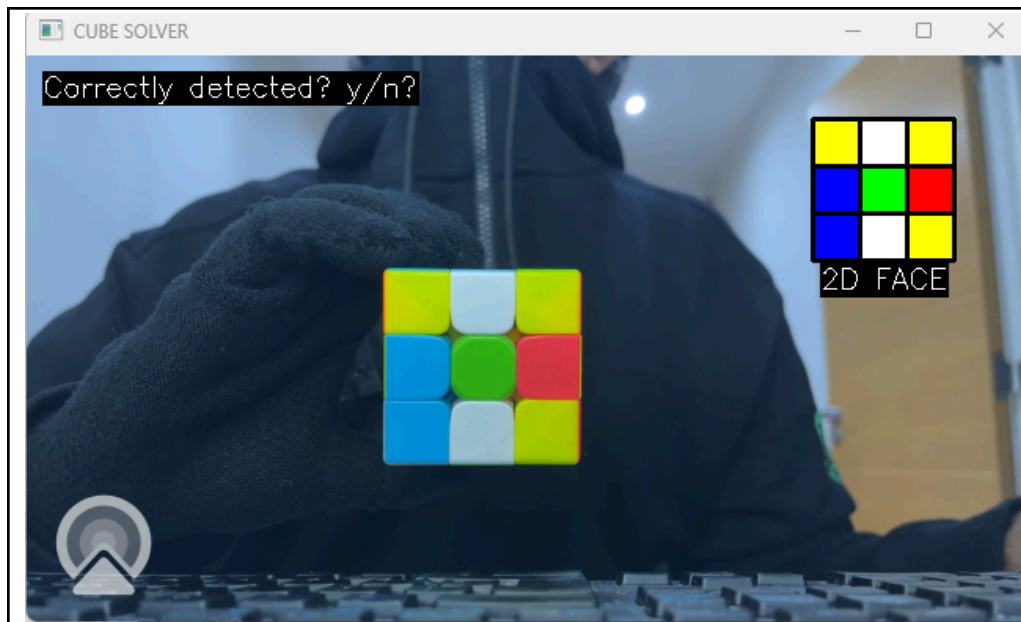
7)The next desired face is the red face, so the augmented arrows are being drawn in the correct direction to guide the user to displaying the red face.



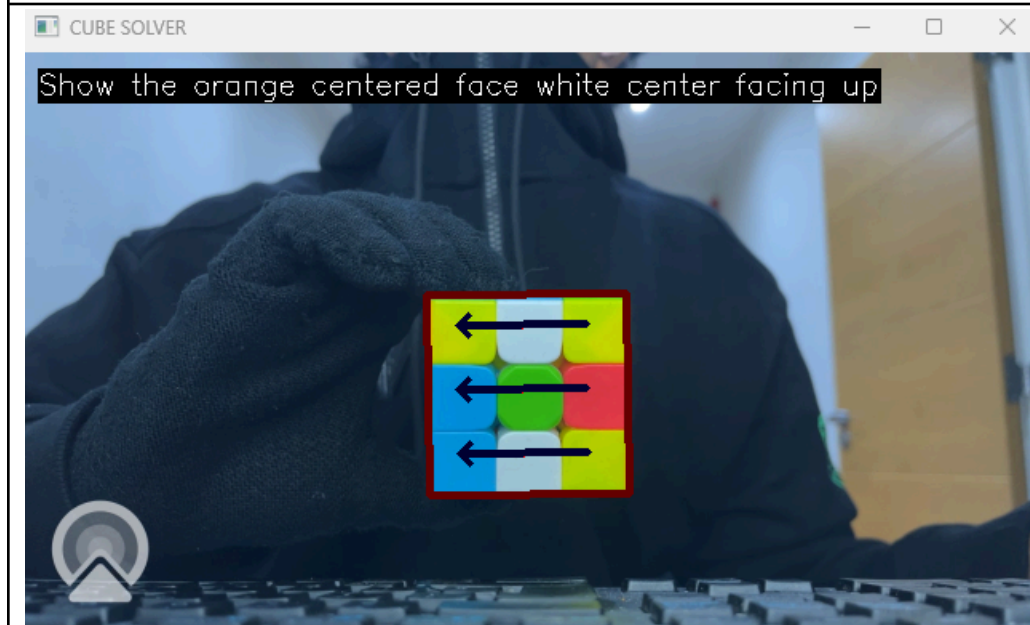
8)The red face is also being correctly displayed and correctly detected on the 2D face.



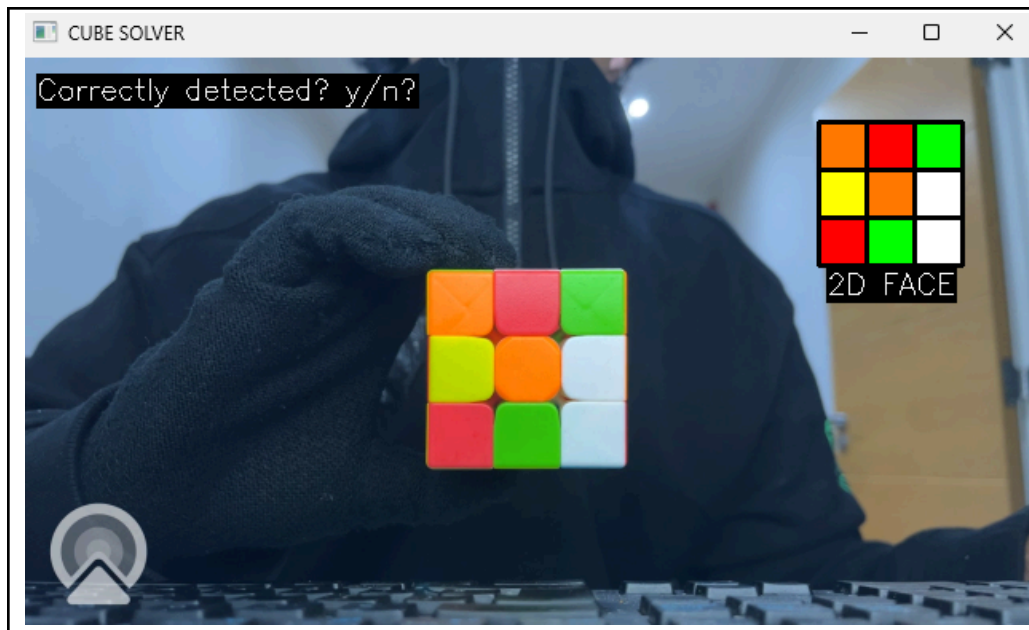
9)The next desired face is the green face, so the augmented arrows are being drawn in the correct direction to guide the user to displaying the green face.



10)The green face is also being correctly displayed and correctly detected on the 2D face.

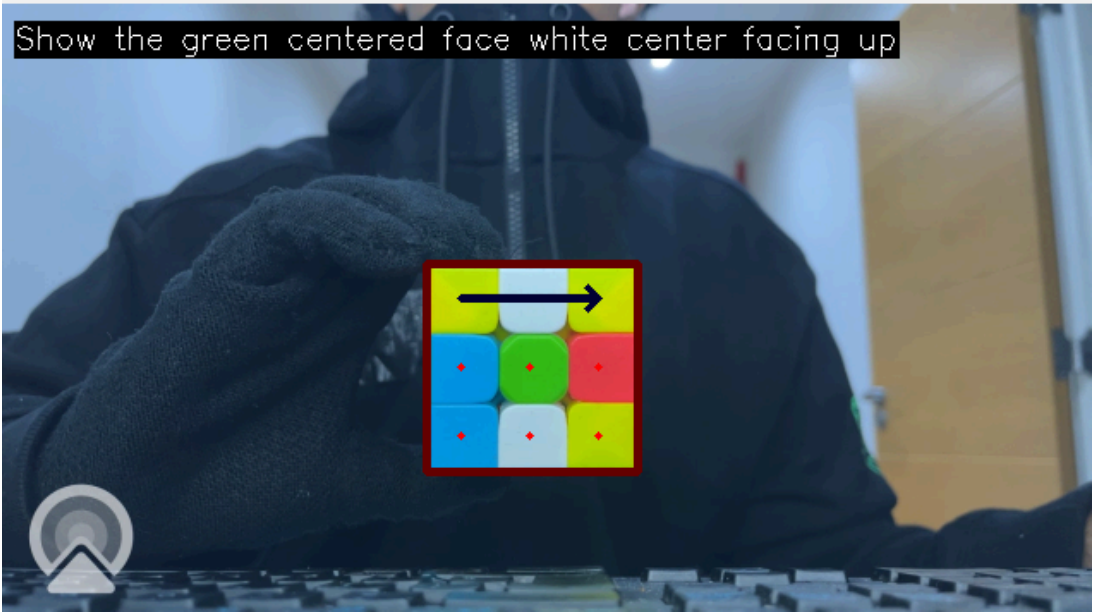


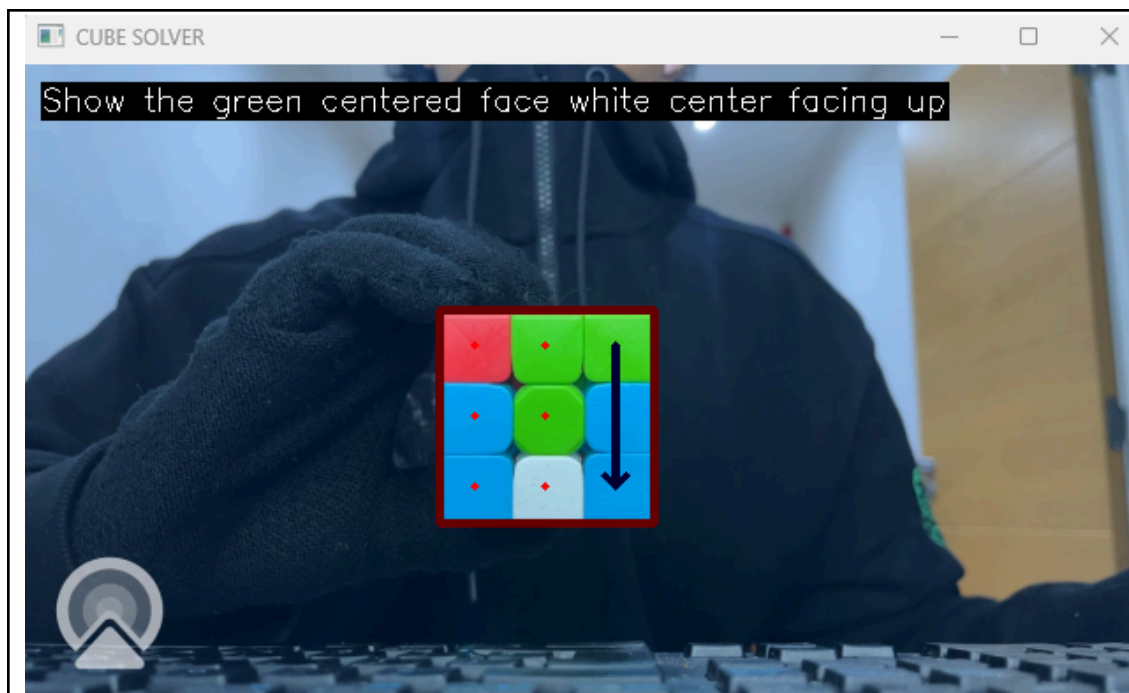
11)The next desired face is the orange face, so the augmented arrows are being drawn in the correct direction to guide the user to displaying the orange face.



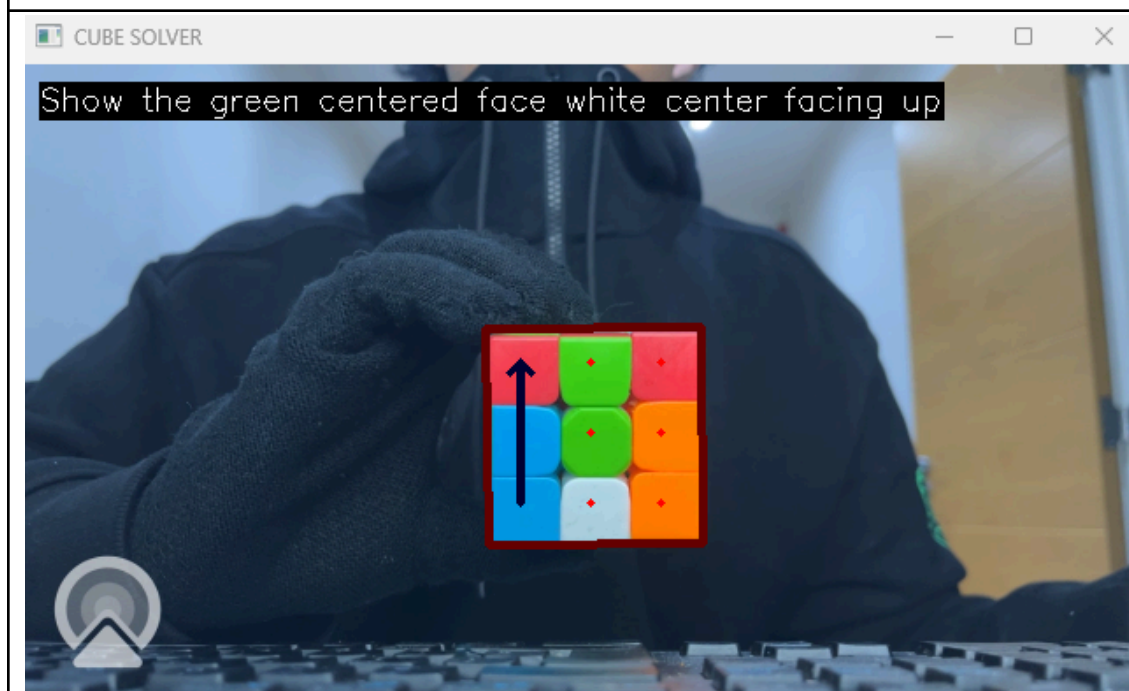
12)The orange face is also being correctly displayed and correctly detected on the 2D face.

Solving the cube:

Frame	Cube move(notation)
CUBE SOLVER Show the green centered face white center facing up 	U'
CUBE SOLVER Show the green centered face white center facing up 	R'



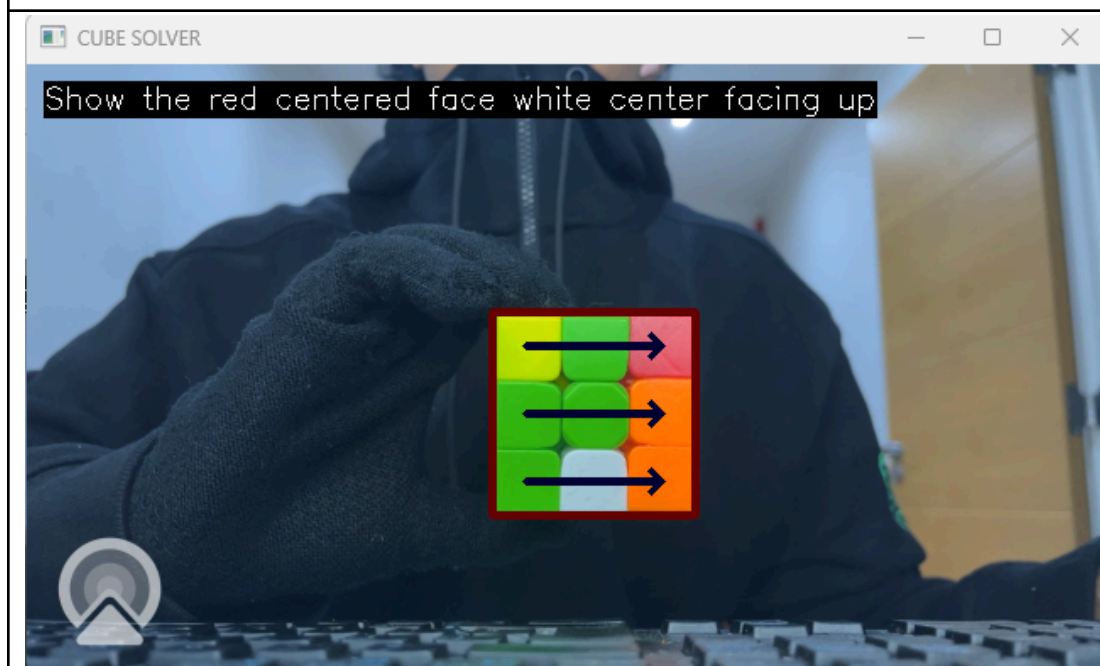
R'



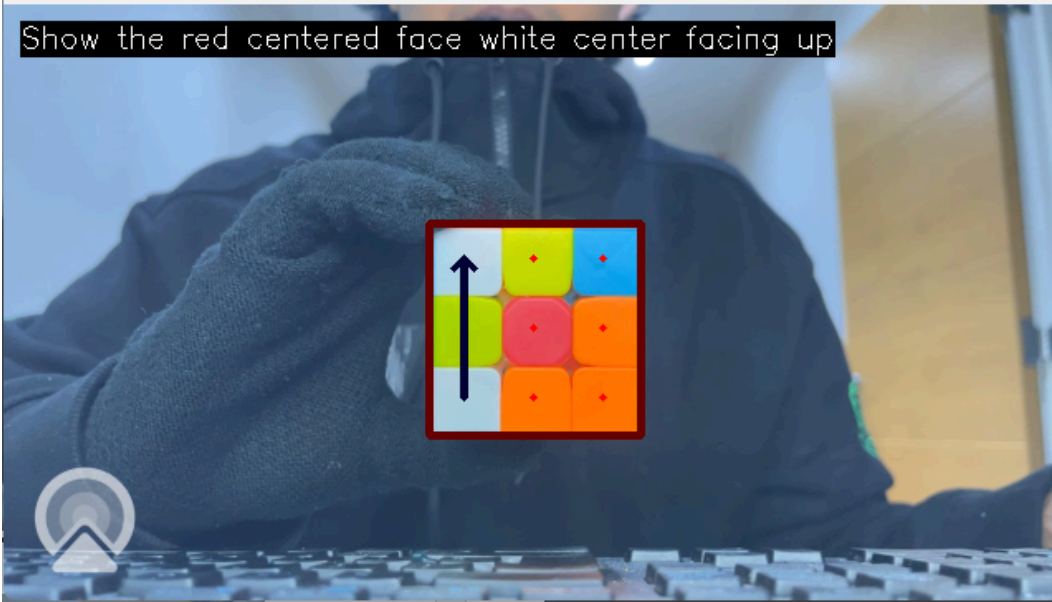
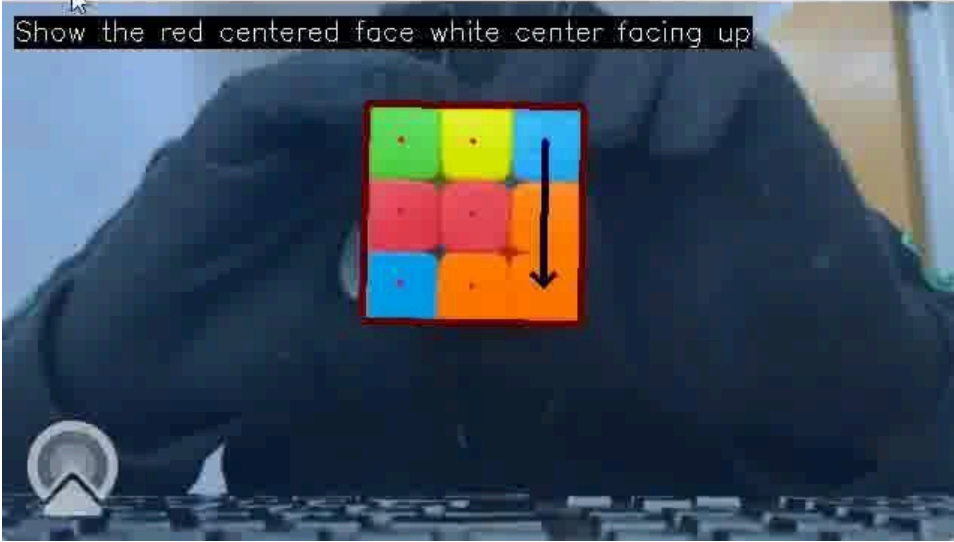
L'

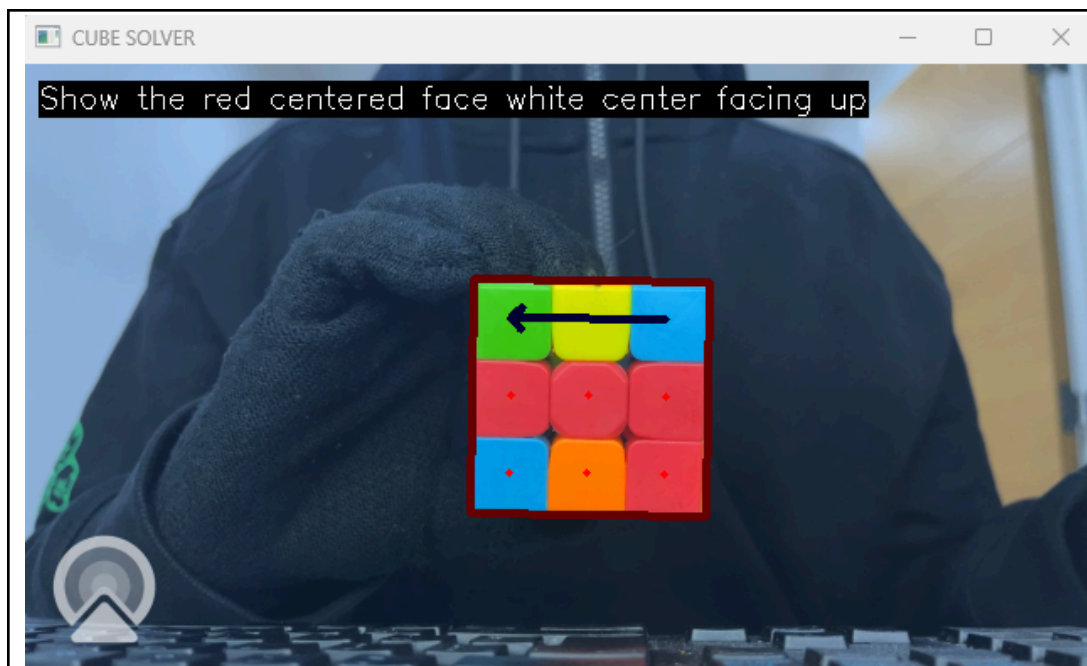


L'

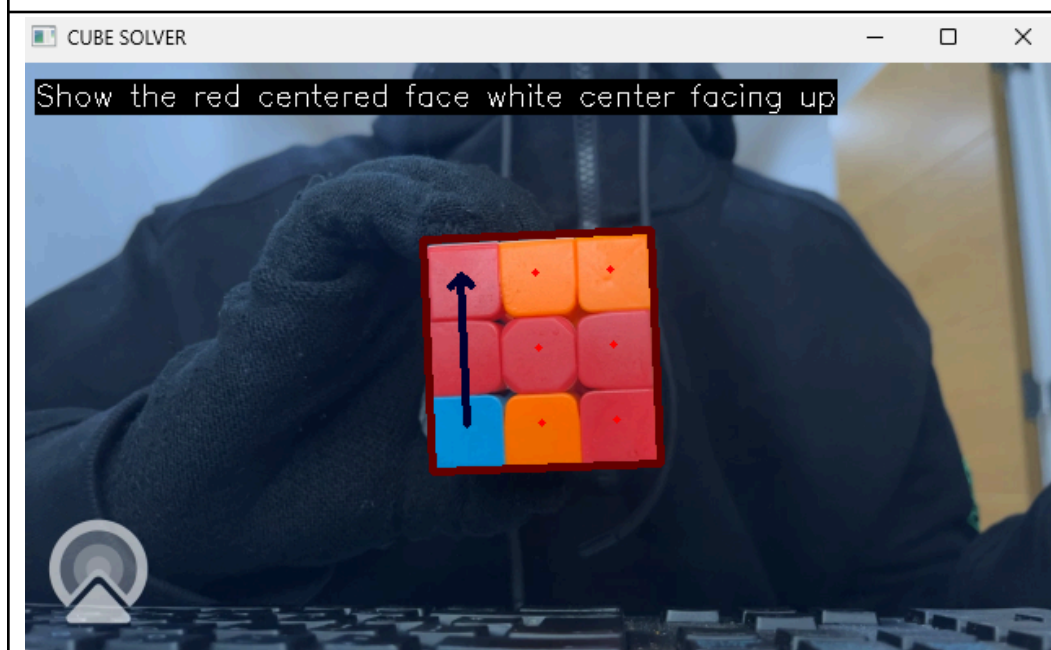


X

 The screenshot shows a window titled "CUBE SOLVER". Below the title bar is a black text box with the instruction "Show the red centered face white center facing up". The main area displays a video feed of a person's hands holding a Rubik's cube. Overlaid on the video is a 3x3 grid representing the cube's face. The grid has a red center square. The top row consists of white, yellow, and blue squares. The middle row consists of yellow, red, and orange squares. The bottom row consists of white, orange, and orange squares. A blue arrow points upwards from the white square in the bottom-left corner of the grid. A small circular icon with a triangle is visible in the bottom-left corner of the video feed.	L'
 The screenshot shows the same "CUBE SOLVER" window. The instruction "Show the red centered face white center facing up" is still present. The video feed shows the same person holding the cube. The 3x3 grid overlay is the same as in the first image, but the blue arrow now points downwards from the blue square in the top-right corner of the grid. The small circular icon in the bottom-left corner of the video feed is now a different icon, resembling a fan or a light.	R'

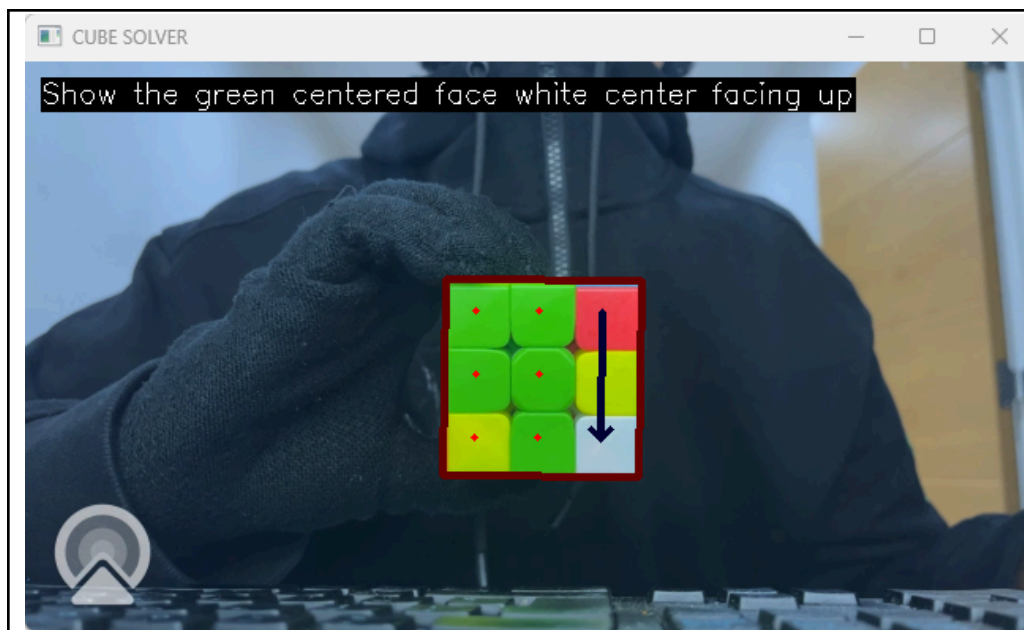


U

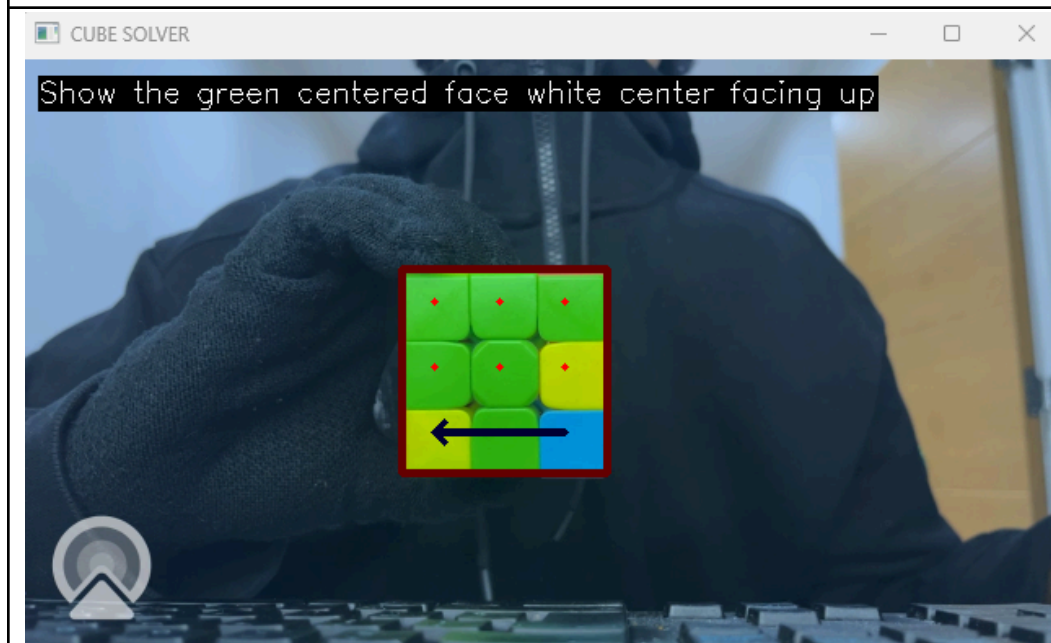


L'

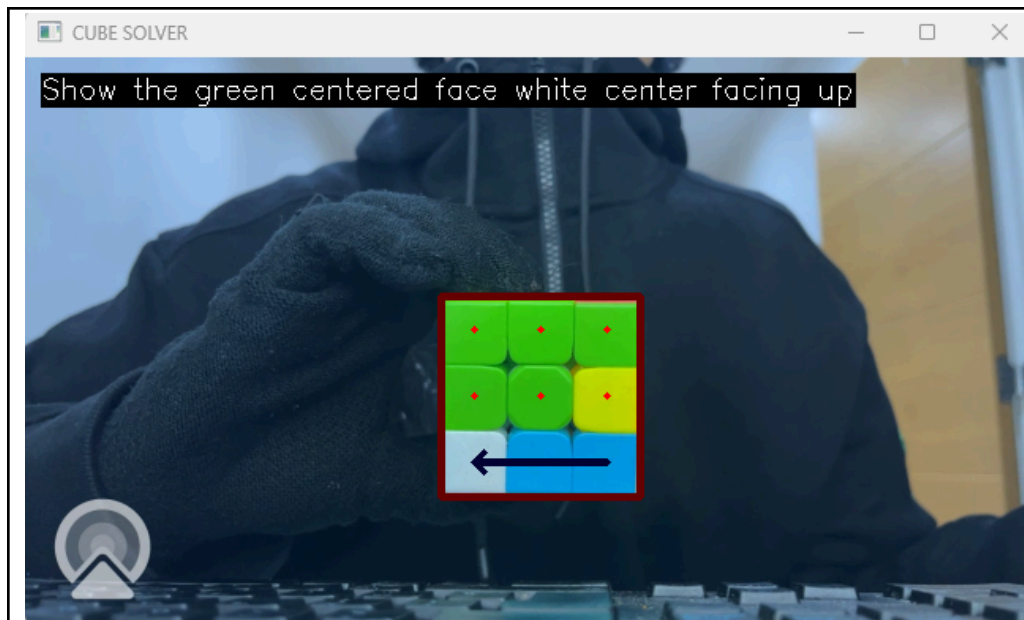
CUBE SOLVER Show the green centered face white center facing up 	
CUBE SOLVER Show the green centered face white center facing up 	R'



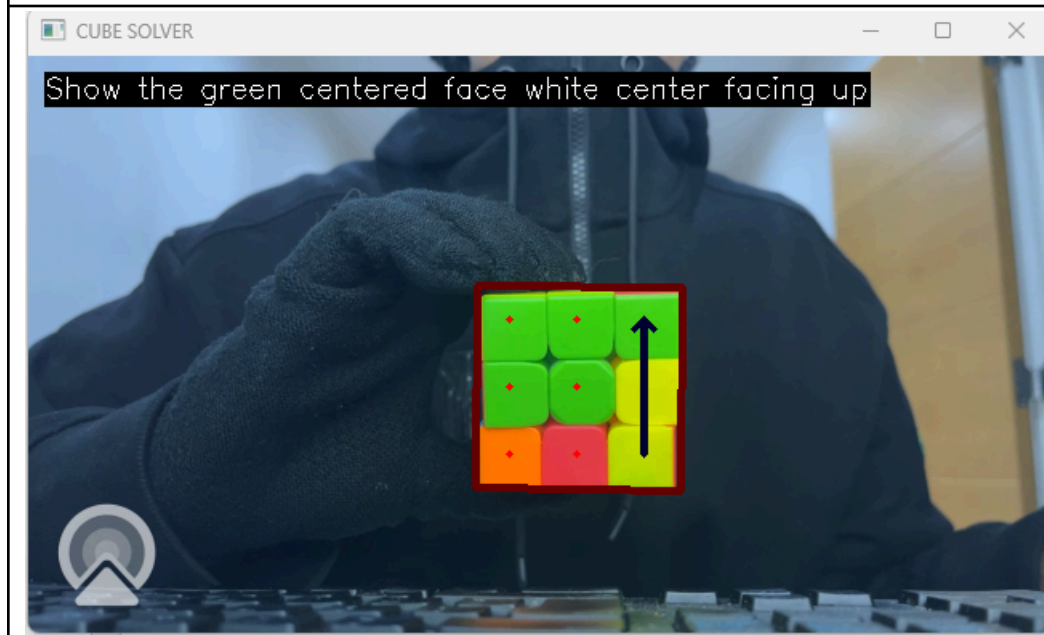
R'



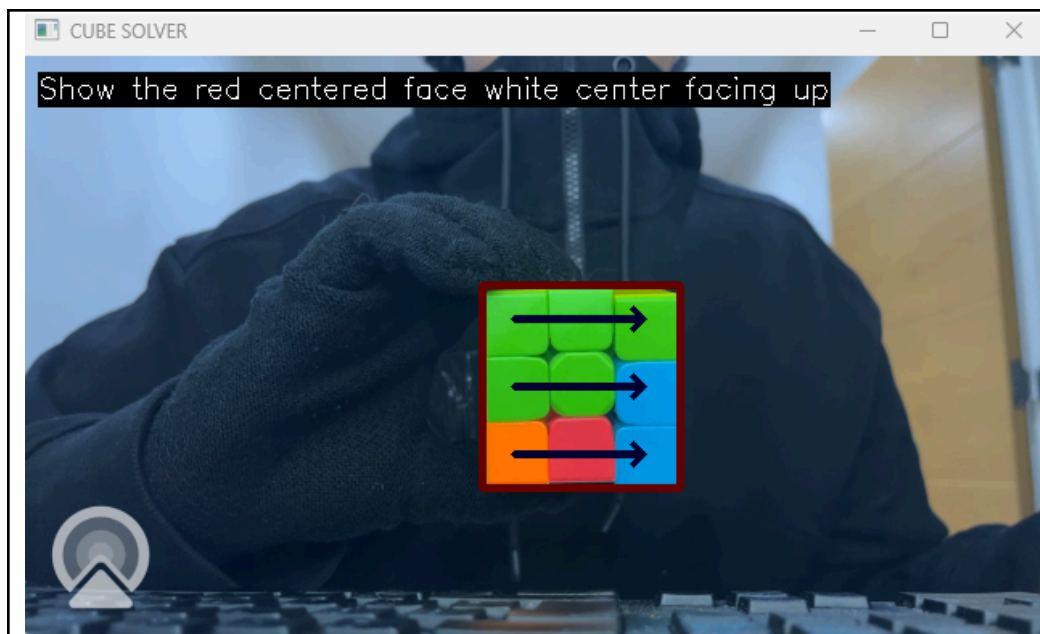
D'



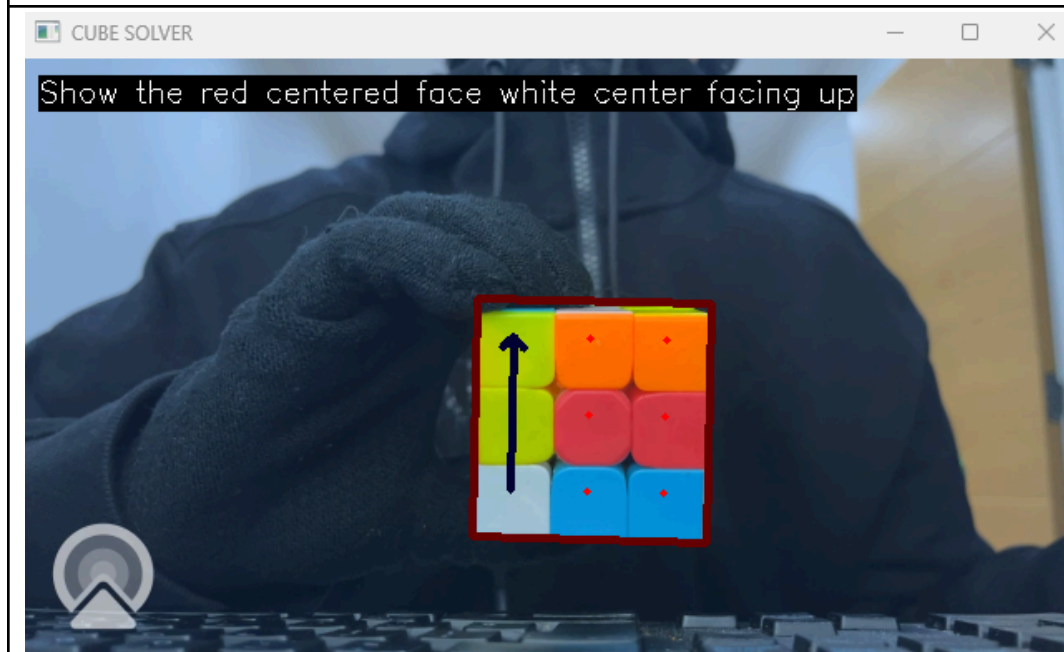
D'



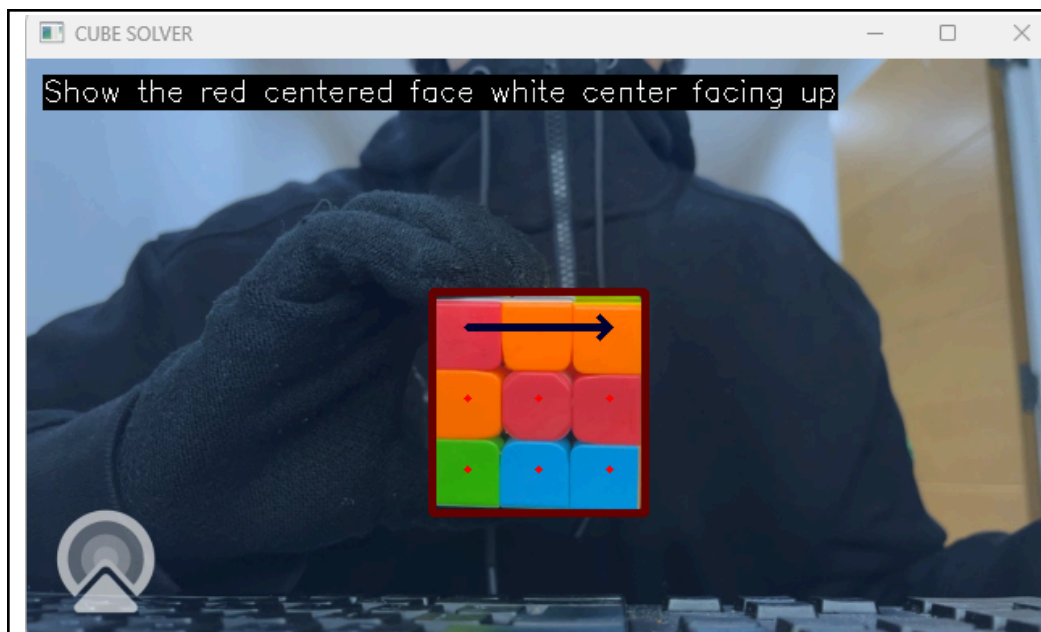
R



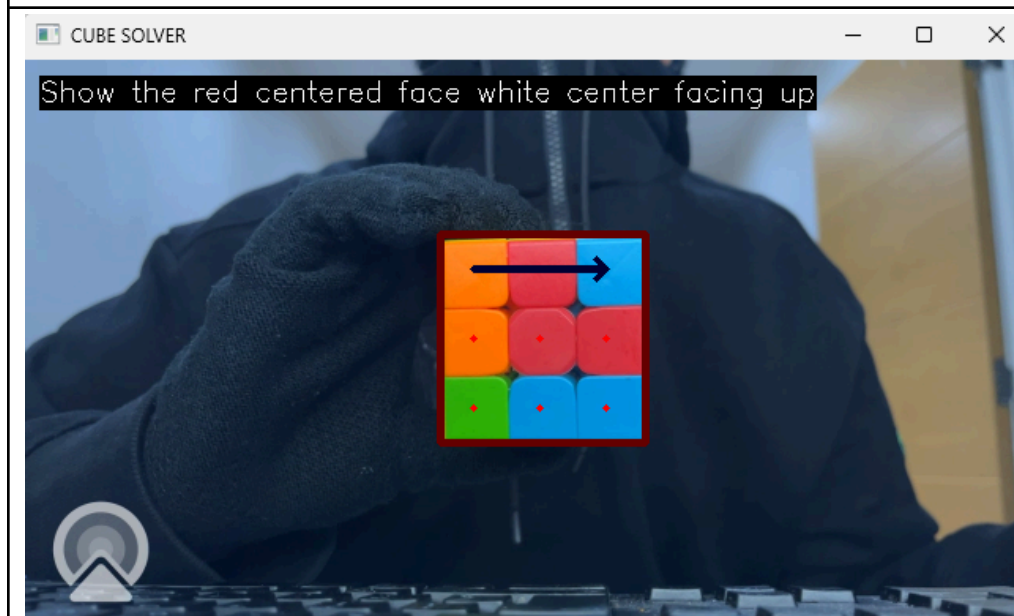
X



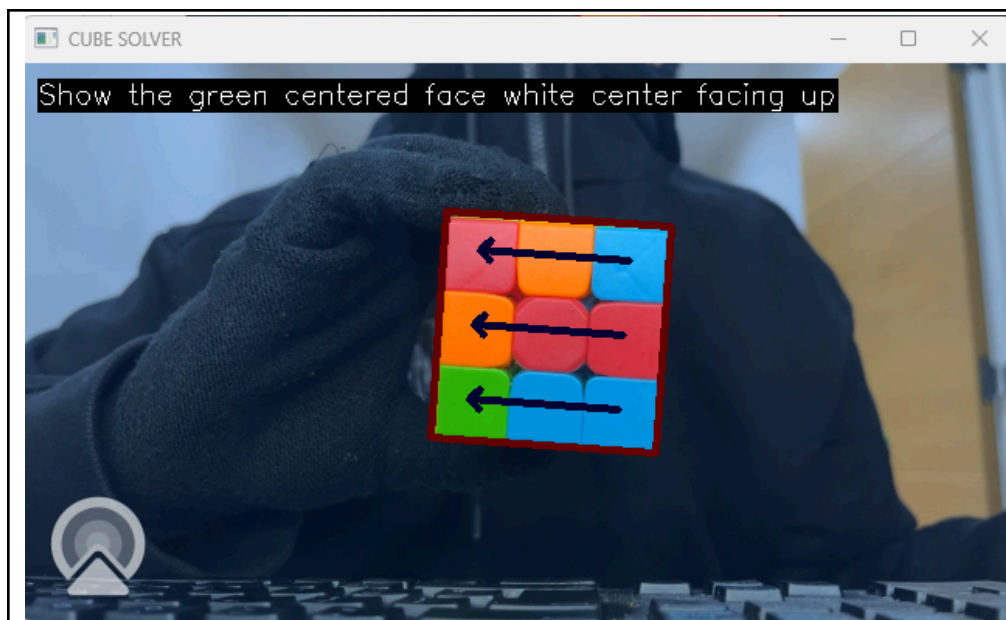
L'



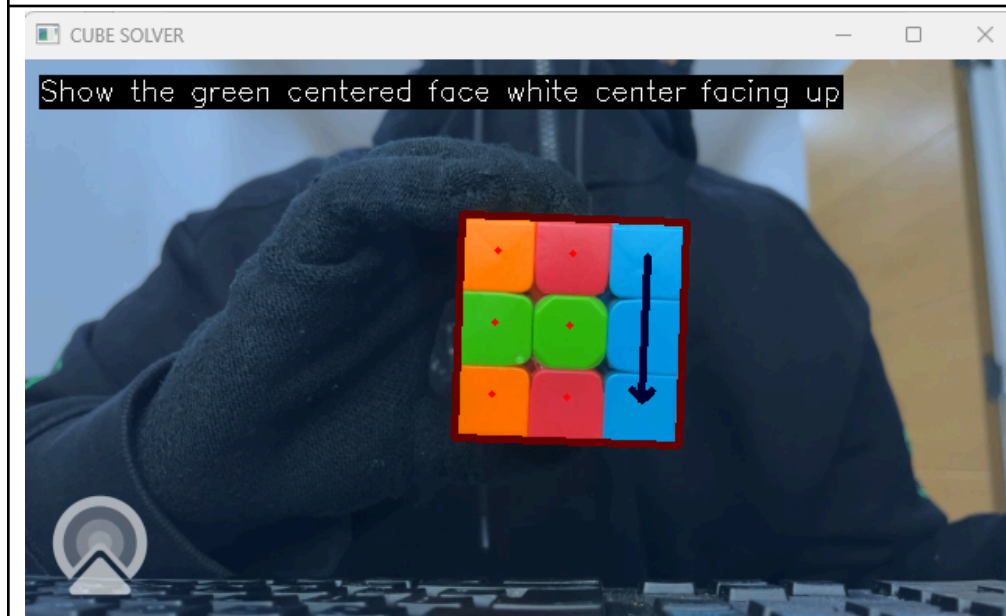
U'



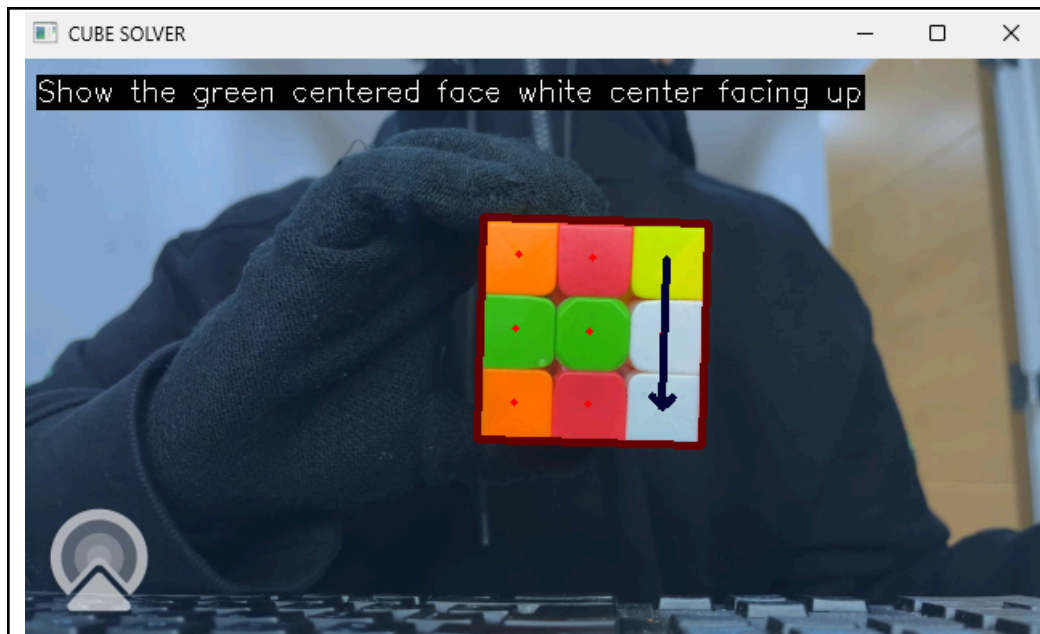
U'



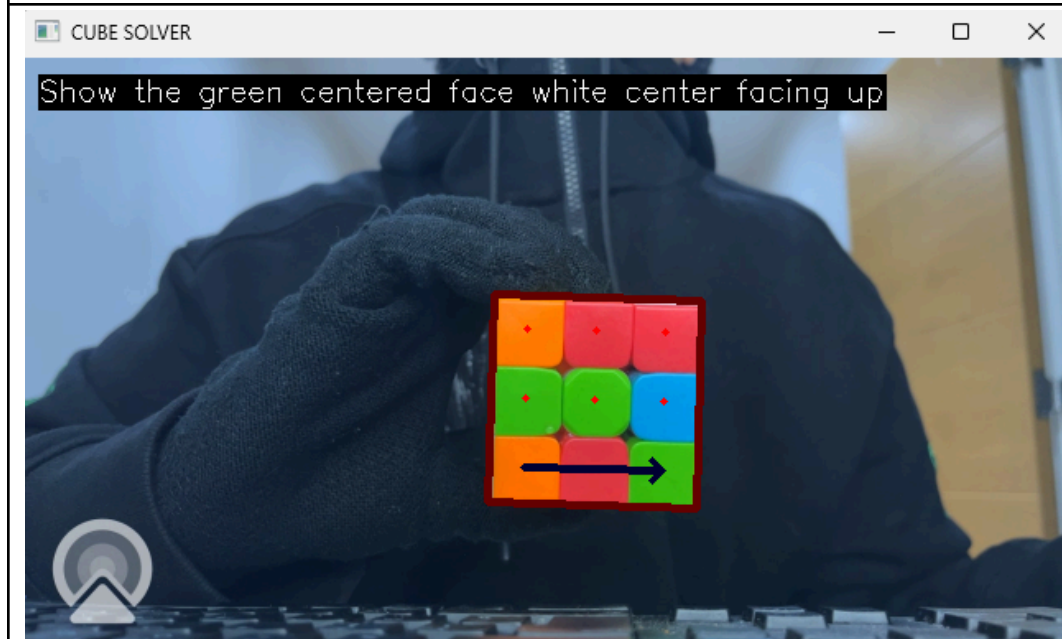
X



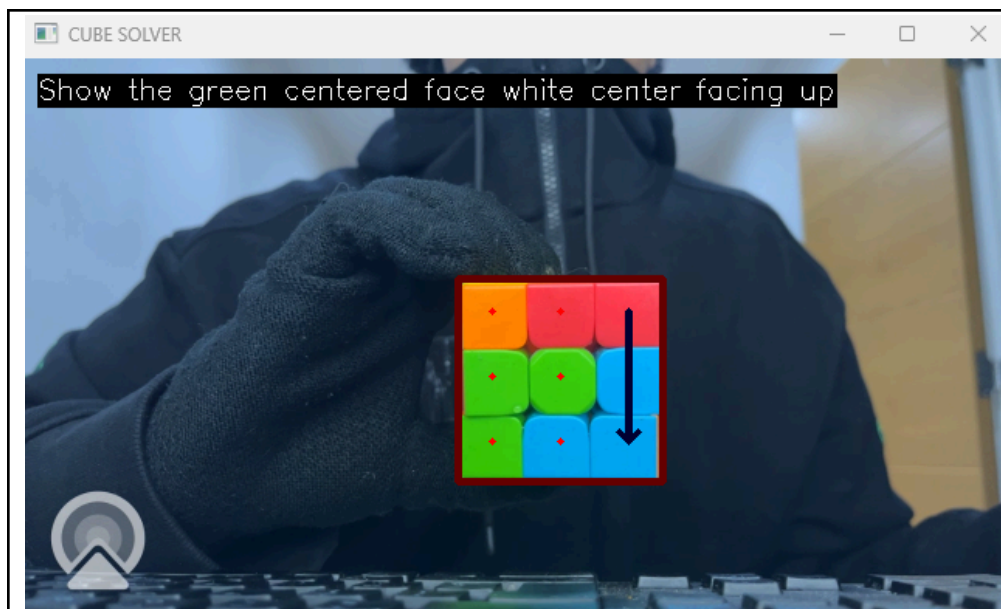
R'



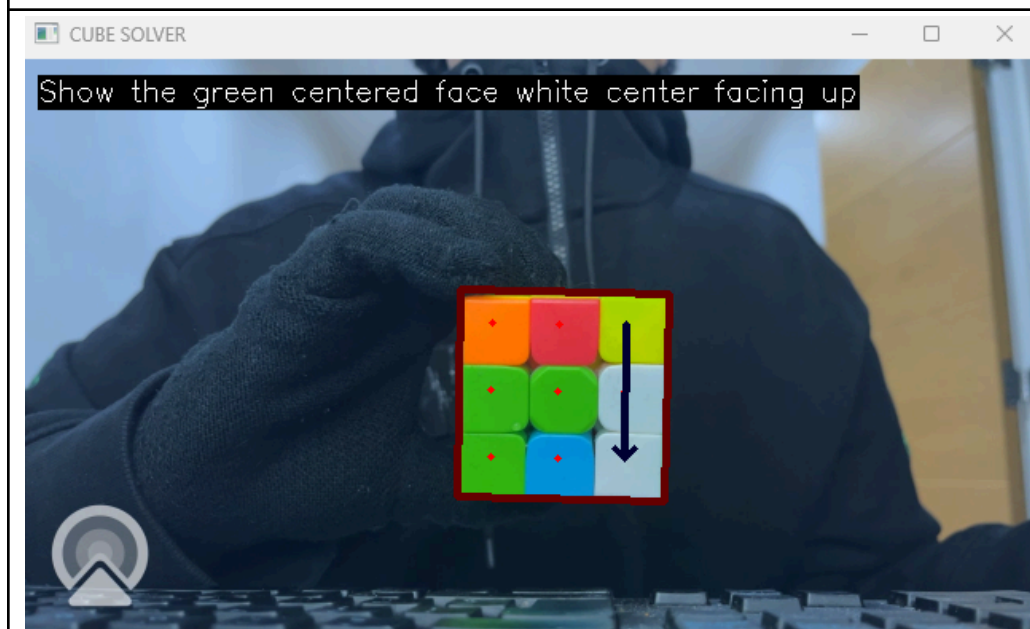
R'



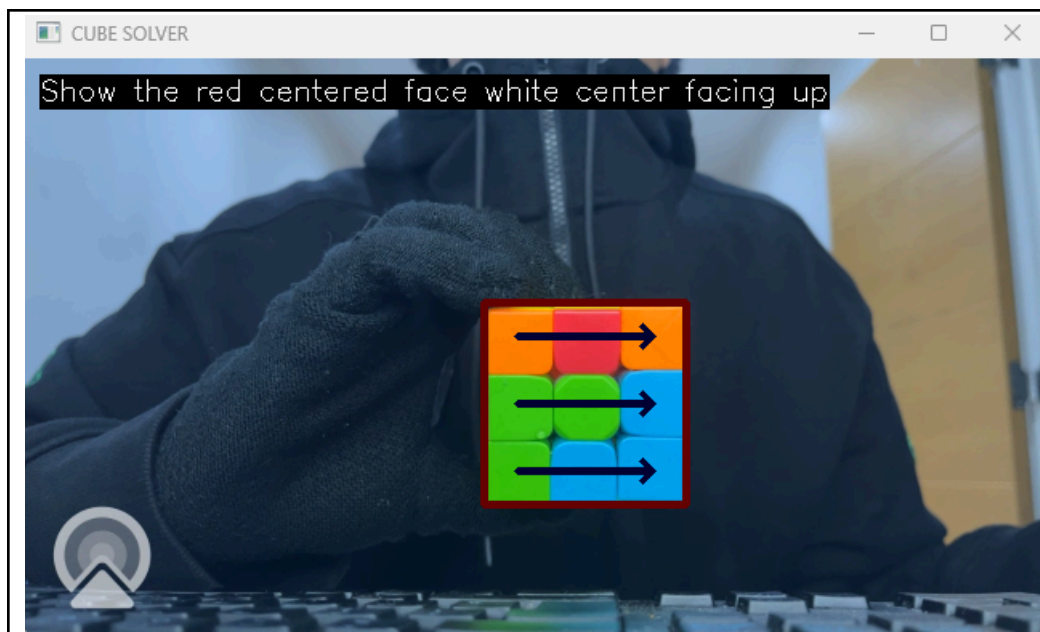
D



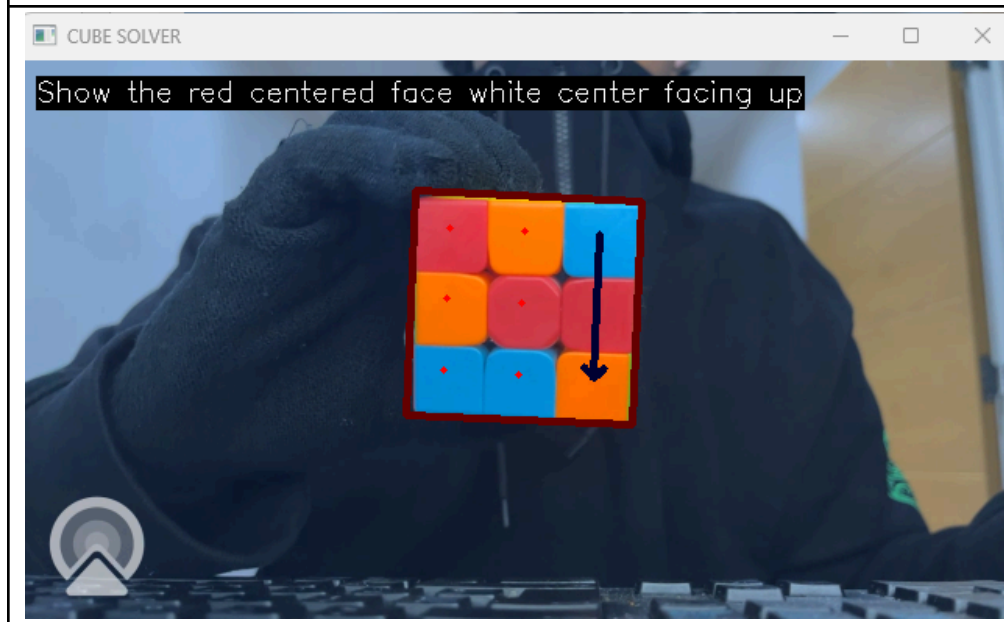
R'



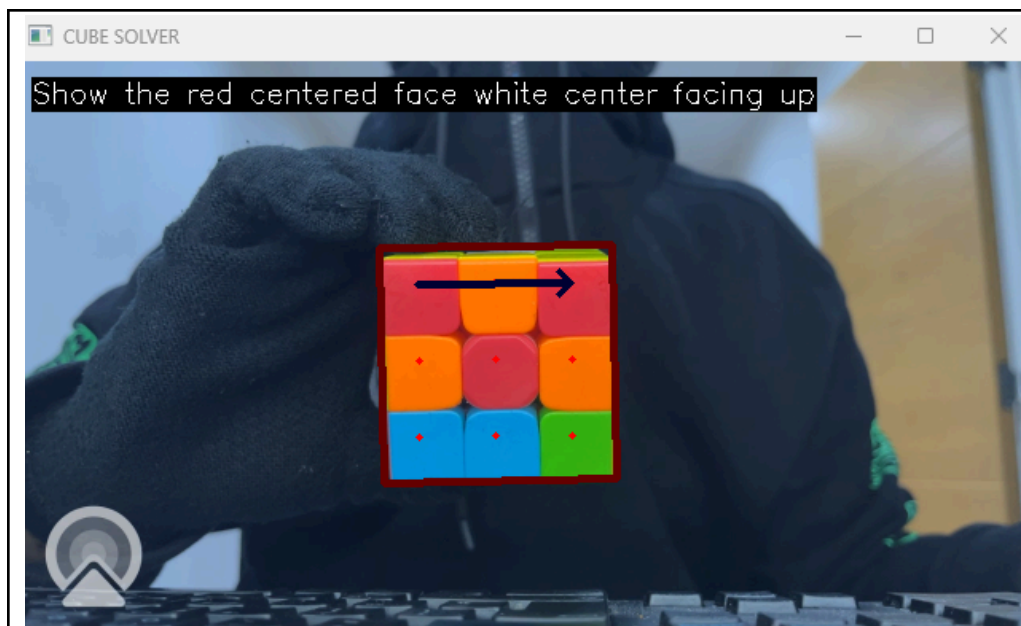
R'



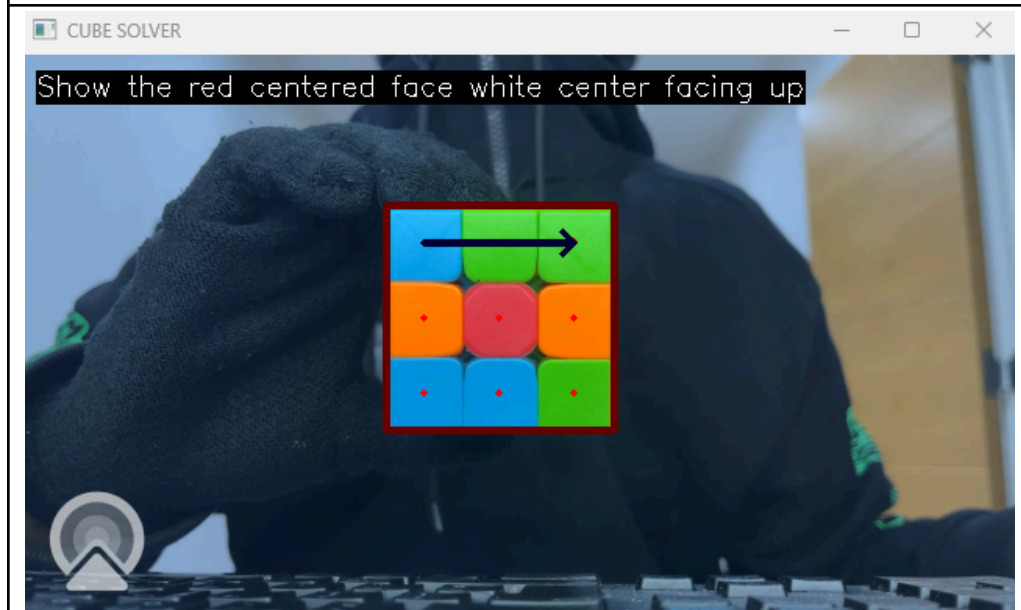
X



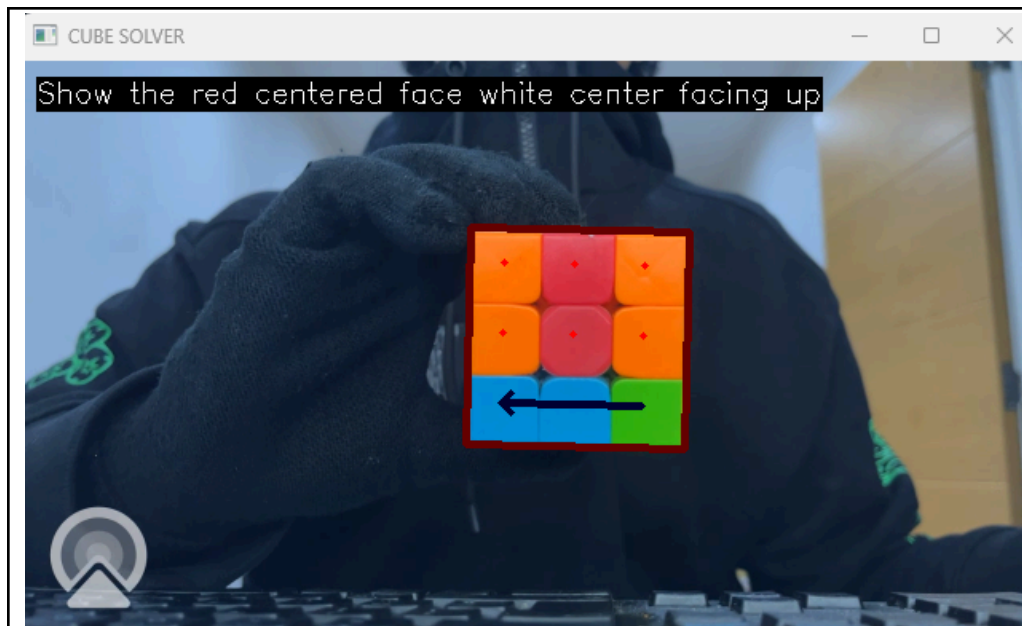
R'



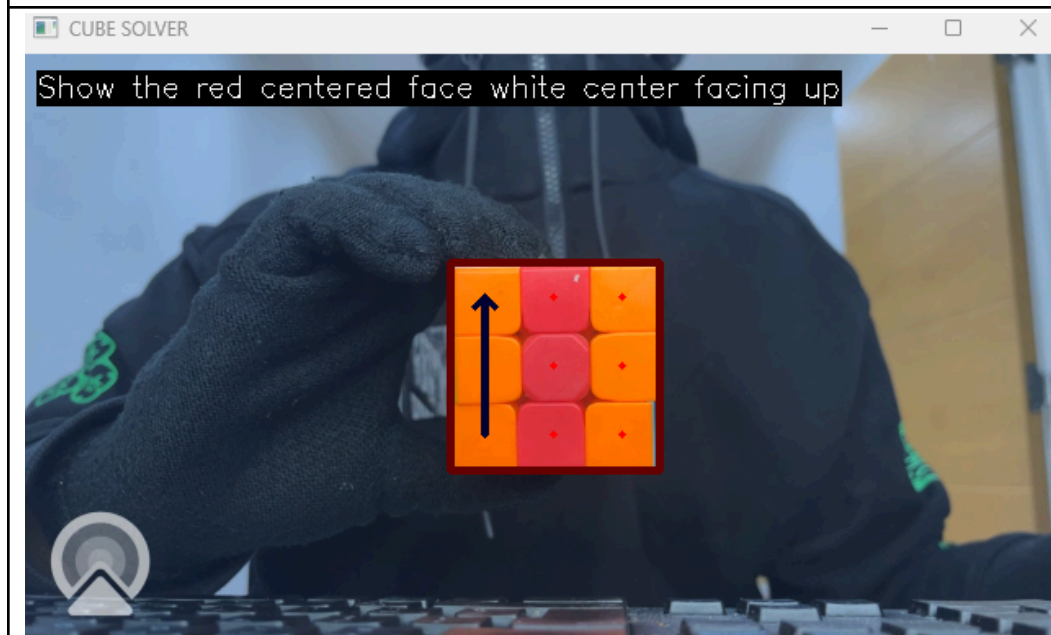
U'



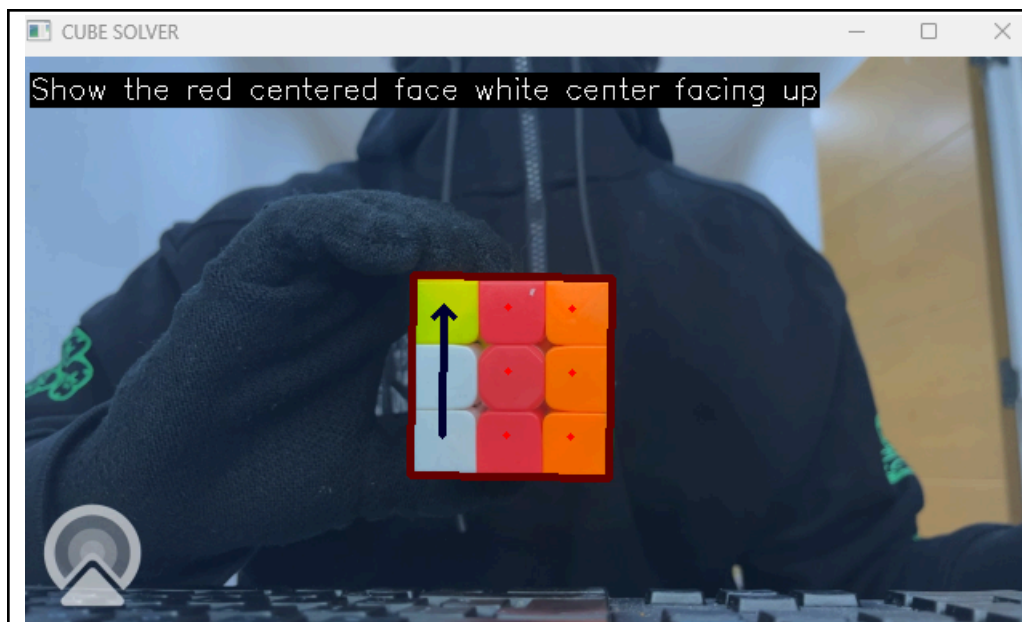
U'



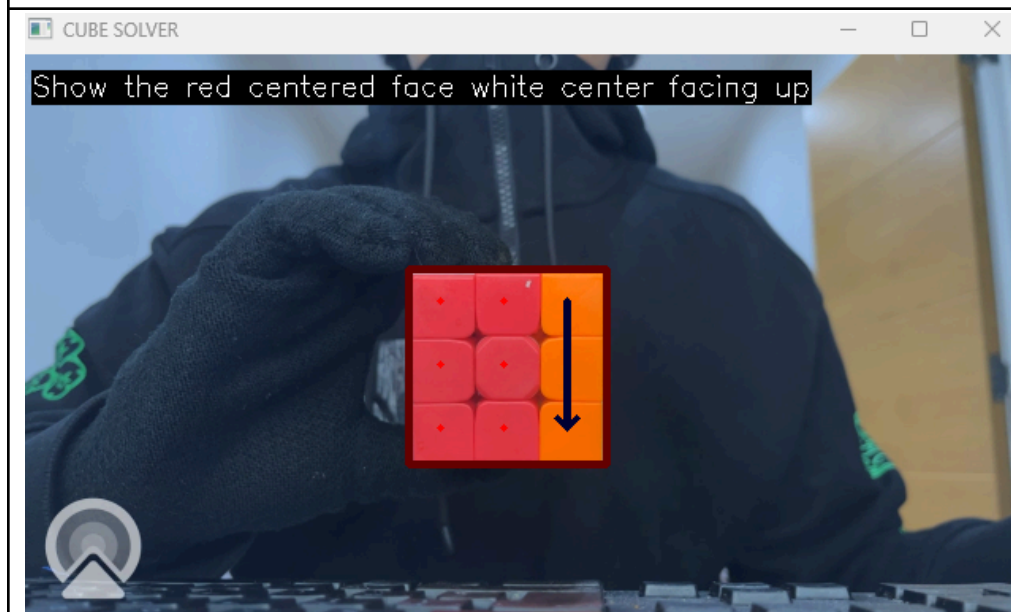
D'



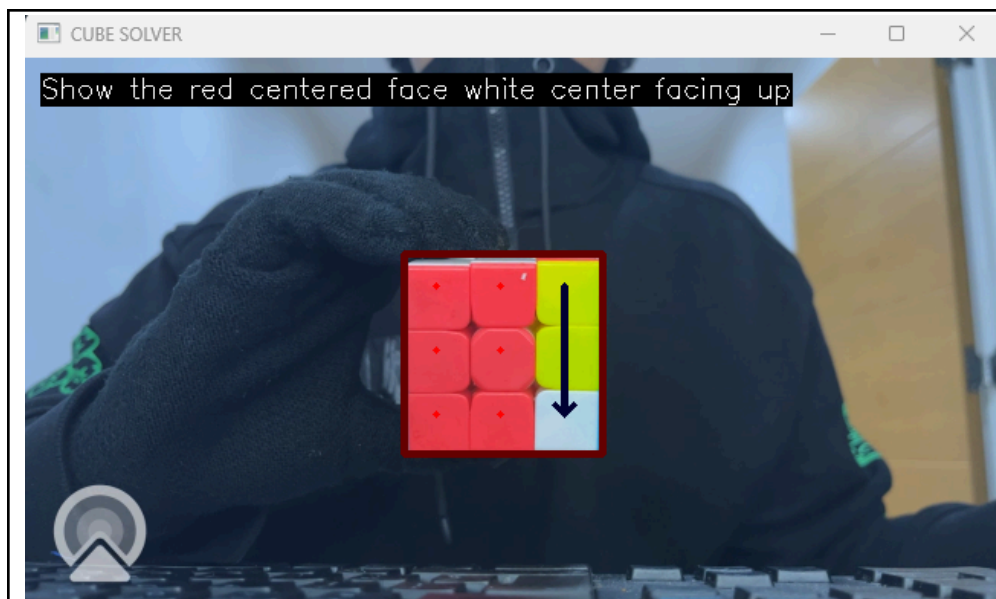
L'



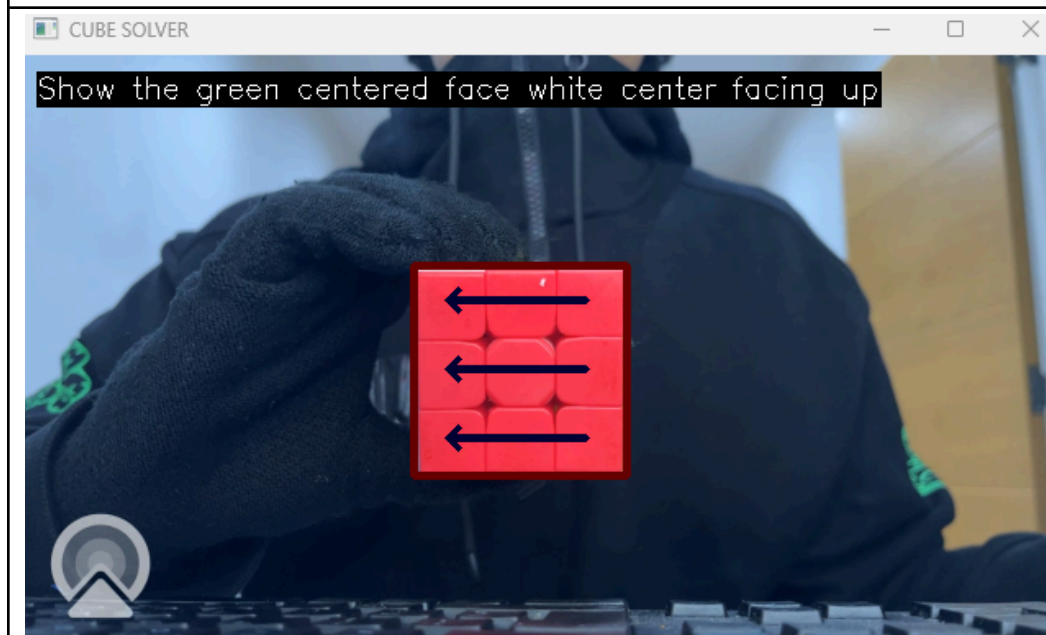
L'



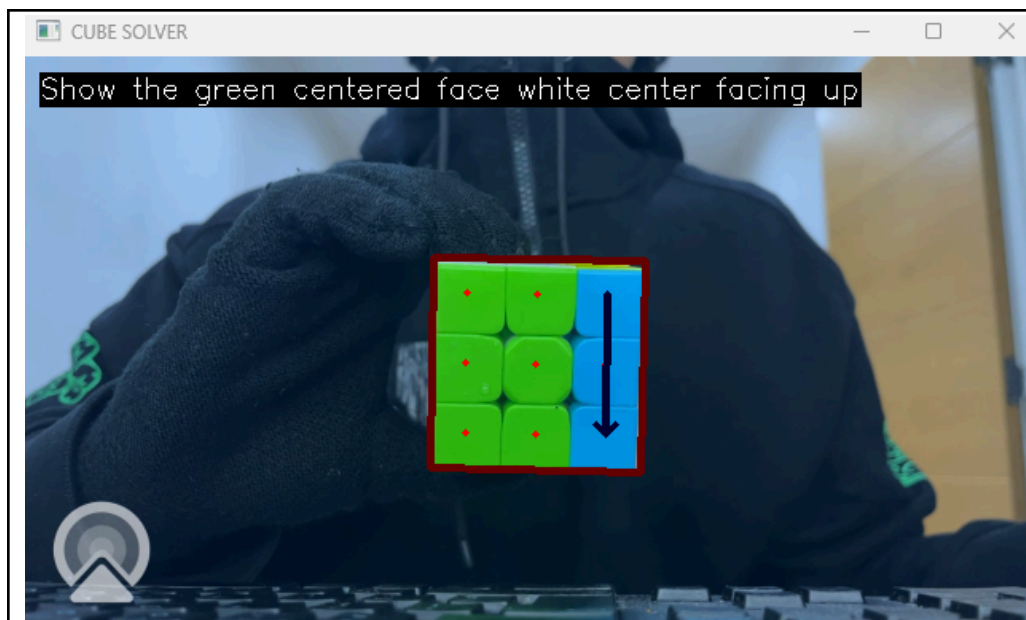
R'



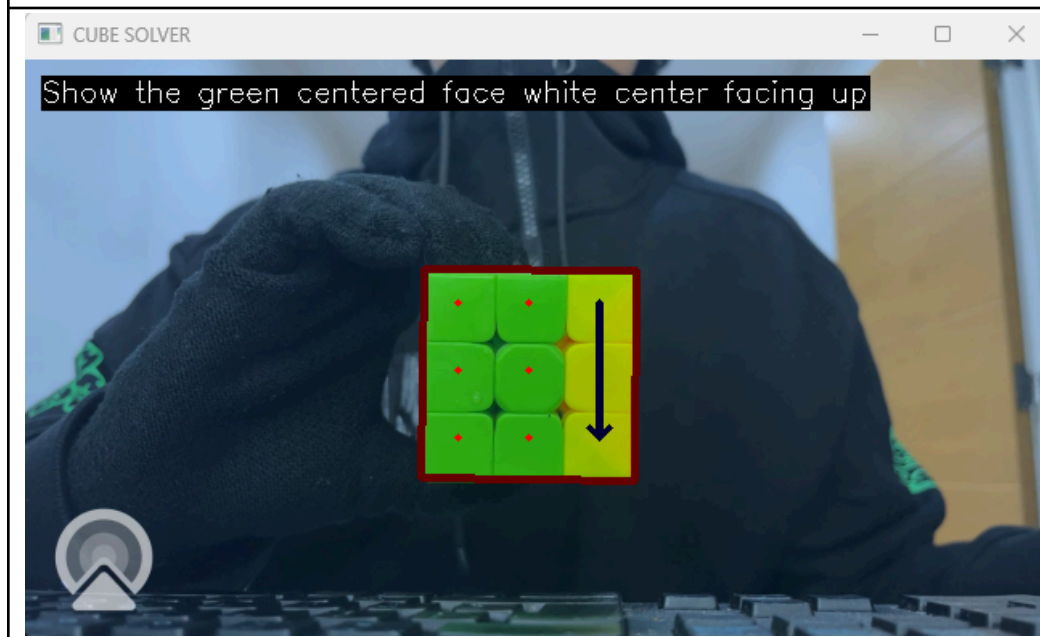
R'



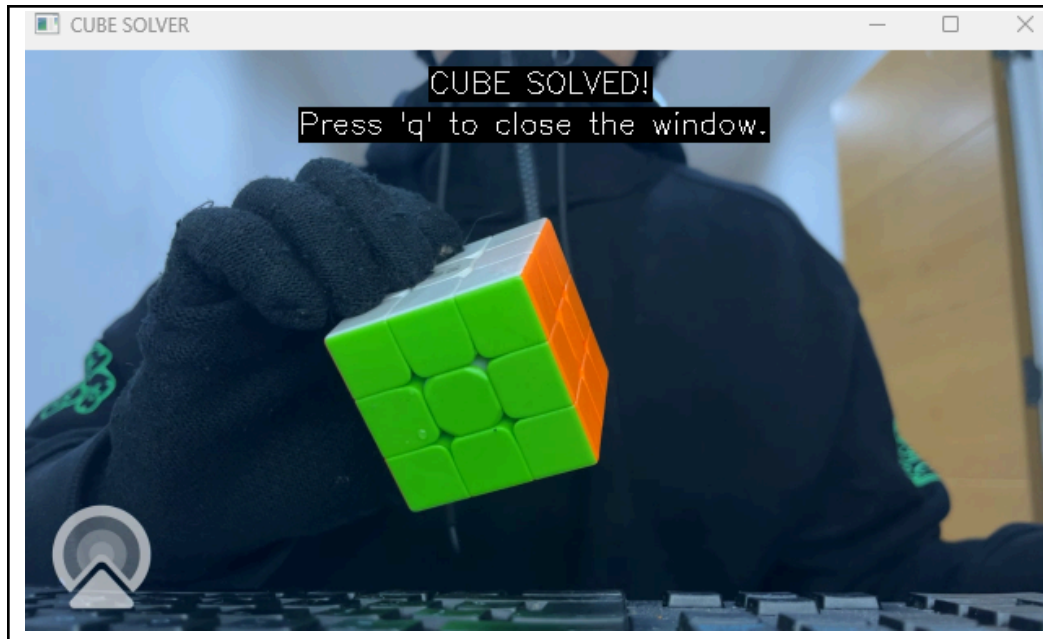
X



R'



R'



The cube has been solved as a result of following the instructions so the application has been successful so far.